

PROJNAME

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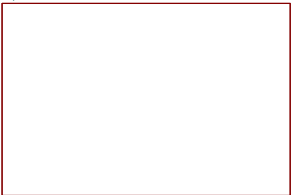
Change history

|       |            |                                                                                                 |
|-------|------------|-------------------------------------------------------------------------------------------------|
| 1.0.0 | 07/10/2026 | First release                                                                                   |
| 1.0.0 | 08/10/2026 | Fix R1234 value due to issue xyz etc etc etc. Update C14 to 10 nF to reduce roll-off frequency. |
| 1.0.0 | TBD        | TBD                                                                                             |
| 1.0.0 | TBD        | TBD                                                                                             |
| 1.0.0 | TBD        | TBD                                                                                             |
| 1.0.0 | TBD        | TBD                                                                                             |
| 1.0.0 | TBD        | TBD                                                                                             |
| 1.0.0 | TBD        | TBD                                                                                             |
| 1.0.0 | TBD        | TBD                                                                                             |
| 1.0.0 | TBD        | TBD                                                                                             |
| 1.0.0 | TBD        | TBD                                                                                             |

ERC FAILURE

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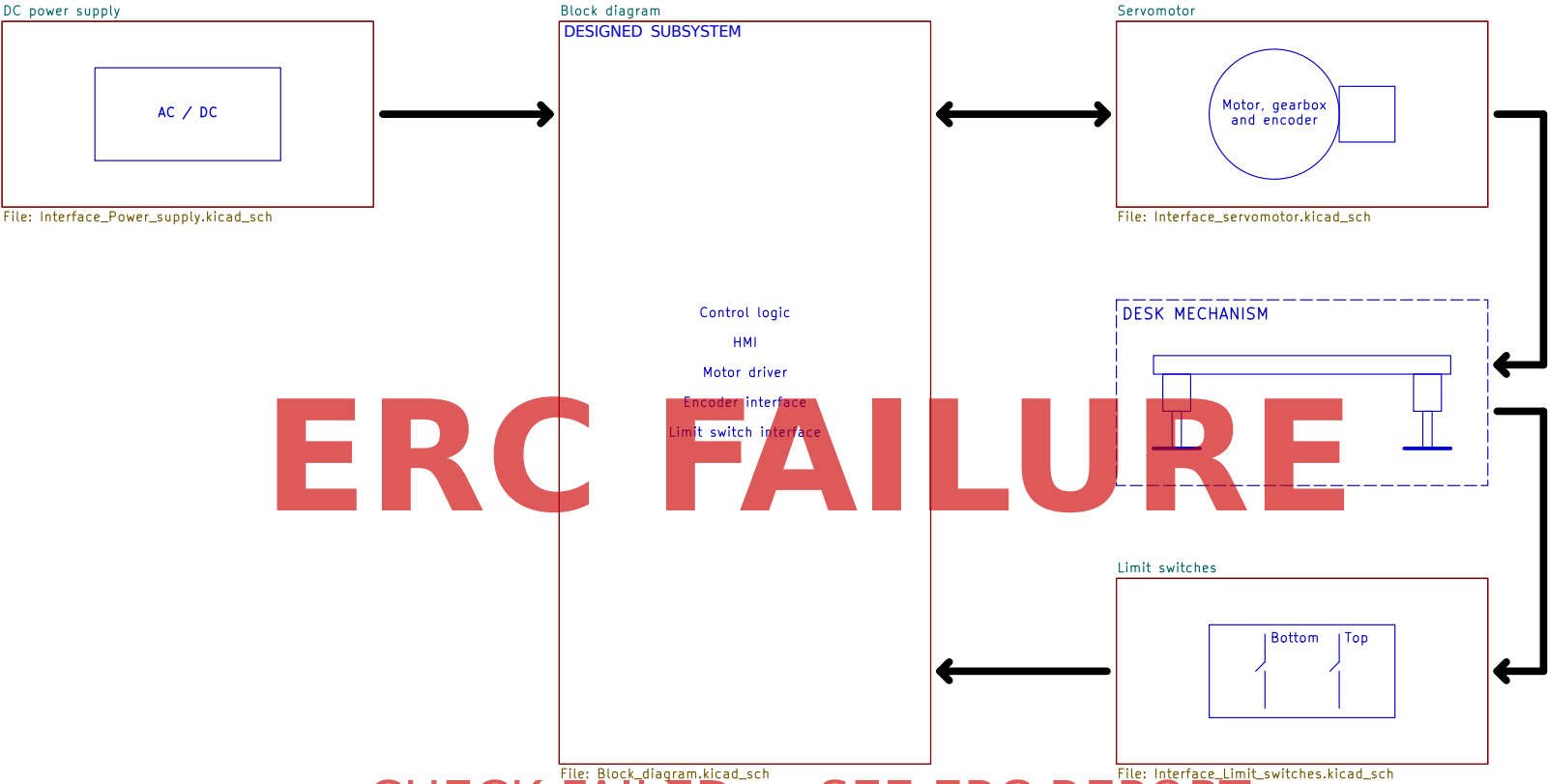
System overview



File: System\_overview.kicad\_sch



|          |                  |                       |            |
|----------|------------------|-----------------------|------------|
| PROJNAME |                  | Sheet: Root           |            |
|          |                  | File: ktest.kicad_sch |            |
| Notes    | Rev: 0.0.1       |                       | Drwn: FLR  |
|          | Mat: Concept     |                       | Rev'd: FLR |
|          | Date: 2026-10-06 |                       | RLs'd: FLR |
|          | Dwg: NA          |                       | Size: A4   |
|          |                  | Generated with KiCad  | Page: 1/6  |



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This sheet shows the overall system and the interfaces to external components.  
External components are shown for system context only and are not part of this schematic.  
Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

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|                      |            |                                 |            |
|----------------------|------------|---------------------------------|------------|
| <b>PROJNAME</b>      |            | Sheet: System overview          |            |
| <b>Notes</b>         |            | File: System_overview.kicad_sch |            |
| Rev:                 | Drwn: FLR  | Mat: Concept                    | Rev'd: FLR |
| Date:                | RLs'd: FLR | Dwg: NA                         | Size: A4   |
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**PROJNAME**

**Notes**

Sheet: Block diagram

File: Block\_diagram.kicad\_sch

Rev: Concept  
Mat: Concept

Date:

Dwg: NA

Generated with KiCad

Drwn: FLR  
Rev'd: FLR  
Rls'd: FLR

Size: A4

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## Interface specification

Barrel connector (Dell type, with ID pin)

Pin

- 1 Supply - 19.5 V, 3 A (inner shell)
- 2 ID (internal pin)
- 3 GND (outer shell)
- SH Shield

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**PROJNAME**

**Notes**

Sheet: DC power supply

File: Interface\_Power\_supply.kicad\_sch

Rev:

Mat: Concept

Date:

Dwg: NA

Generated with KiCad

Drwn: FLR

Rev'd: FLR

RLs'd: FLR

Size: A4

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Interface specification

JST B6B-XH-A / JST XHP-6 connector

| Pin | Description            |
|-----|------------------------|
| 1   | Encoder B output (VCC) |
| 2   | Encoder A output (VCC) |
| 3   | GND                    |
| 4   | VCC (3.3 - 5V)         |
| 5   | Motor B (3 A, 12-24 V) |
| 6   | Motor A (3 A, 12-24 V) |

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|          |                                      |            |
|----------|--------------------------------------|------------|
| PROJNAME | Sheet: Servomotor                    |            |
|          | File: Interface_servomotor.kicad_sch |            |
|          | Rev:                                 | Drwn: FLR  |
|          | Mat: Concept                         | Rev'd: FLR |
|          | Date:                                | RLs'd: FLR |
| Notes    | Dwg: NA                              | Size: A4   |
|          | Generated with KiCad                 | Page: 11/6 |

Interface specification

JST B4B XH A / JST XHP-4 connector

- Pin Description
- 1 Bottom limit switch (Normally open, switch closes to GND)
  - 2 GND
  - 3 GND
  - 4 Top limit switch (Normally open, switch closes to GND)

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|          |                                          |            |
|----------|------------------------------------------|------------|
| PROJNAME | Sheet: Limit switches                    |            |
|          | File: Interface_Limit_switches.kicad_sch |            |
|          | Rev:                                     | Drwn: FLR  |
|          | Mat: Concept                             | Rev'd: FLR |
|          | Date:                                    | RLs'd: FLR |
| Notes    | Dwg: NA                                  | Size: A4   |
|          | Generated with KiCad                     | Page: 12/6 |